Nanyang Technological University
School of Electrical & Electronic Engineering
IE2006 Engineering Mathematics I

**Tutorial 8 Numerical Methods**

1. Use the method of **fixed-point iteration** to solve the following equations. Give your answers correct to 3 decimal places.

   (Go through the steps given in the lecture notes.)

   (a) $x^5 = x + 0.12, \quad x \in [1, 1.5]$

   (Hint: take $g(x) = (x + 0.12)^{1/5}$)

   (b) $x^2 e^x = 1, \quad x \in [0.5, 1]$

   (Hint: take $g(x) = e^{-x/2}$)

2. Given the following table

| $x$ | 0.2 | 0.4 | 0.5 | 0.7 |
|---|---|---|---|---|
| $\sin x$ | 0.198669 | ?? | 0.479426 | 0.644218 |

   Find

   (i) the **Newton's divided difference interpolating polynomial**
   (ii) the missing value (??)
   (iii) an upper estimate for the error.

3. Given the following table

| $x$ | 1.5 | 1.7 | 1.9 | 2.0 | 2.1 |
|---|---|---|---|---|---|
| $x^{1/3}$ | 1.144714 | 1.193483 | 1.238562 | ?? | 1.280579 |

   Find

   (i) the **Newton's divided difference interpolating polynomial**
   (ii) the missing value (??)
   (iii) an upper estimate for the error.

Answers

1. (a) 1.028 (b) 0.703

2. (i) $0.198669 + (x - 0.2)(0.935857) + (x - 0.2)(x - 0.5)(-0.223793)$

   (ii) 0.390316 (iii) $0.98 \times 10^{-3}$

3. (i) $1.144714 + (x - 1.5)(0.243845) + (x - 1.5)(x - 1.7)(-0.046125) + (x - 1.5)(x - 1.7)(x - 1.9)(0.013083)$

   (ii) 1.259914 (iii) $0.14 \times 10^{-4}$